

Lesson 7: Project - Designing an App Part 2

135 minutes

Overview

This lesson covers the last three days of the unit project. Students add code to make their apps function.

- On Day 3, students learn about Pair Programming by watching a video and then practicing it themselves while working on their project apps. At this stage, students are adding their first lines of code to their app using debugging skills from the previous lesson.
- On Day 4, students continue working on their apps. Halfway through class the focus of the lesson shifts to getting feedback. Students watch other groups test their apps and collect feedback that will be used to make updates.
- On Day 5, students complete their apps, making any final adjustments based on feedback from their peers. Students spend some time reviewing other apps that classmates made and then complete a short set of reflection prompts before submitting their projects.

Standards

Full Course Alignment

CSP Conceptual Framework

- ▶ **CRD-1** - Incorporating multiple perspectives through collaboration improves the computing innovations being developed.
- ▶ **CRD-2** - Developers create and innovate using an iterative design process that is user-focused, that incorporates implementation/feedback cycles, and that leaves ample room for experimentation and risk-taking.

CSTA K-12 Computer Science Standards (2017)

- ▶ **AP** - Algorithms & Programming

Agenda

Day 3 (45 minutes)

Warm Up

Activity

Wrap Up

Day 4 (45 minutes)

Objectives

Students will be able to:

- Create the code and user interface of an app based on a program specification
- Effectively use pair programming while designing the features of an app
- Iteratively improve an app based on feedback
- Provide effective feedback on the functionality or usability of an app
- Reflect on the value of different stages of a development process in creating an app
- Test an app's functionality by attempting to use features and behavior described in a program specification

Preparation

- Make sure students have access to their App Development Planning Guides
- Check the "[Teacher's Lounge](#)" forum for verified teachers to find additional strategies or resources shared by fellow teachers
- If you are teaching virtually, consider checking our Virtual Lesson Modifications for [Day 3](#), [Day 4](#), and [Day 5](#).

Links

Heads Up! Please make a copy of any documents you plan to share with students.

For the teachers

Warm Up

Activity

Wrap Up

Day 5 (45 minutes)

Warm Up

Activity

Wrap Up

- **CSP Bingo Cards - Unit Three** - Handout ([Download](#))
- **CSP Bingo Generator** - Resource
- **Designing an App - Days 3-5** - Slides
[▼ Make a Copy](#)
- **Writing Support - Unit 3 Project - Designing an App** - Handout
[▼ Make a Copy](#)

For the students

- **App Development Planning Guide** - Activity Guide [▼ Make a Copy](#)
- **Pair Programming** - Video ([Download](#))

Vocabulary

- **Development Process** - the steps or phases used to create a piece of software. Typical phases include investigating, designing, prototyping, and testing.
- **Pair Programming** - a collaborative programming style in which two programmers switch between the roles of writing code and tracking or planning high level progress

Teaching Guide

Day 3 (45 minutes)

Warm Up (5 minutes)

Important context for teachers: Students will continue to work on adding behavior and code to their app in following two days. We recommend you take a look at Day 4 and Day 5 as well when you are preparing for this lesson to get an idea of where students should be at the end of this lesson and what tasks and activities take place over the course of the last three days of the project.

What Makes a Good Partner?

 **Discuss:** *What makes a good partner?'*

Discussion Goal: Make this a quick discussion and just aim to get a few ideas from around the room. For example, a good partner...

- listens
- contributes
- shares the work evenly
- and so on....

Remarks

Now you've got the skills you'll need for your project, but we'll need a minute to talk about how to use them when working in teams. Today you're mostly going to have work time, but you're going to be practicing a new skill called pair programming.

Activity (35 minutes)

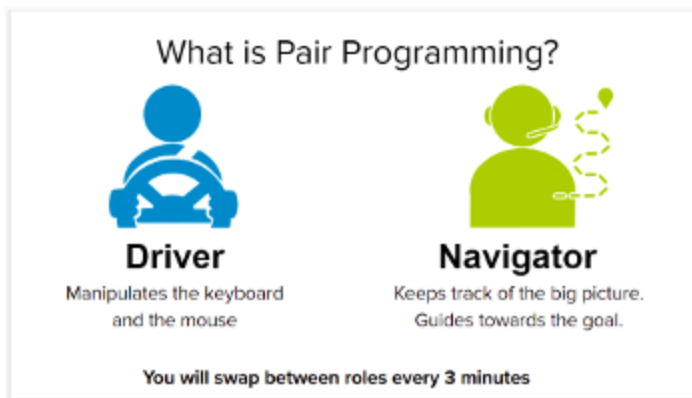
Using Pair Programming

Group: Place students in pairs with their project partner

Teaching Tip

Taking on the Blank Screen: This is the first time students are adding code to a blank screen. Some students may need assistance. Direct them back to their Planning Guides to help students understand the **onEvent** blocks they need to add to their project.

Supporting Pair Programming: Your biggest role in supporting pair programming will be encouraging students to use it early on and helping enforce the switching. Run the timer in the slides and have left and right partners switch roles. You may opt to increase the time to longer periods as time goes on.



Debugging Practices: Students learned a lot of debugging practices in the previous lesson. Encourage students to use these and record bugs they find as they program today. Continue to normalize and celebrate debugging as a normal and fun part of programming.

Distribute: Direct students back to the [App Development Planning Guide](#)

Step 6: Students fill out the chart on page 4, listing all of the Event Handlers in their programs. They should be able to determine these based on the Program Specification they designed.

Display: Play the Pair Programming video found in [r cspunit3-introtoapp_design/csp/2025]. Afterwards review the pair programming steps found in Code Studio.

Remarks

Today we're going to use the pair programming as you work on your app with your partner. To begin we'll make the partner sitting on the left the driver and the one on the right the navigator. Every few minutes I'll ask you to switch roles.

Circulate: Give students time to work on their project. As they do so circulate the room encouraging them to use debugging practices they've learned in previous lessons. Every few minutes ask left and right partners to switch being drivers and navigators.

Wrap Up (5 minutes)

Reflecting on Pair Programming

 **Discuss:** *How does Pair Programming help when working on a project? How does it help with the debugging process in particular?*

Discussion Goal: Use this prompt to reinforce not only the value of pair programming and collaboration while programming, but also to remind students of debugging practices they learned in the previous lesson. Possible answers include

- Talking through my code with another person helps me figure out what to do
- When I get stuck, there's someone there to help brainstorm a solution
- There's more than one set of eyes to find bugs
- Different partners bring different perspectives to a project

 **Journal:** Have students add "Pair Programming" to their journals.

Day 4 (45 minutes)

Warm Up (5 minutes)

 **Discuss:** *Think of times when you've received helpful feedback on school work, a hobby, or a sport.*

- *What makes good feedback?*
- *What makes bad feedback?*

Discussion Goal: This is an optional prompt if your class situation requires it. If you believe your students need more time to work on their apps then move right into work time. This prompt, however is designed to get students thinking early about the differences between useful and unhelpful feedback. For example, students may note that...

- Good feedback:
 - Specific information beyond "it's great".
 - Explains why something needs work
- Bad Feedback:
 - Is overly negative without constructive feedback
 - Does not go into enough detail

Remarks

Today we're going to keep working on our apps, but we're also going to be getting feedback from our classmates. Let's get to it!

Activity (30 minutes)

Group: Place students back in pairs with their project partners.

Work Time - 20 minutes

🎤 Remarks

For the next twenty minutes, you and your partner will use Pair Programming to continue working on your app. After this is done, you will get feedback from another group.

Circulate: Direct students to Code Studio to continue working on their apps.

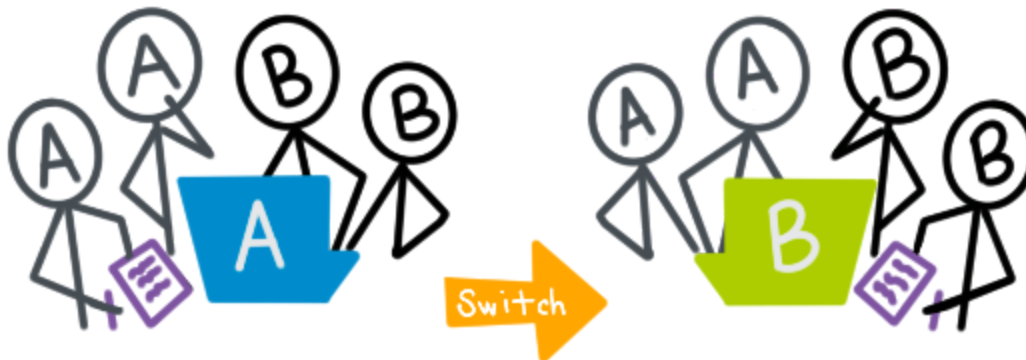
Testing and Feedback - 10 minutes

📋 **Distribute:** Direct students back to the [App Development Planning Guide](#).

📋 **Step 7:** Groups swap apps and gather feedback. It is ok to move ahead to this section even if students are not done with their apps yet.

Here's what you should observe:

- Group A lets Group B test their app.
- Group A watches Group B use the app, and writes down improvements that come to mind.
- Group A interviews Group B asking for specific improvements Group A can make. This may be different than what Group A came up with during the observation. These improvements are written down in the Activity Guide.
- Group A and B swap and now Group A tests Group B's app while Group B watches.
- After this, mix up the group pairings and repeat the steps above.



🖥 2

Add More Code

Wrap Up (10 minutes)

🎤 Remarks

📋 **Step 8:** Now return to your stations and pick at least one improvement to make to their app based on feedback. Write this down in the Planning Guide in Step 7.

📋 **Discuss:** *Why is it important to get feedback from others while building your app? What is the value of getting this feedback even if you aren't finished with your app?*

Discussion Goal: Use this prompt as time allows. The goal is to reinforce the importance of feedback in the development of good software. Possible answers include:

- Another person may find a bug that you have been overlooking
- Something that seems obvious to you is not obvious to the user
- Even if the app isn't finished, feedback can help guide the next steps

Remarks

Feedback and testing are important parts of making good software. They ensure that our apps actually work the way we designed them, and they help us make gradual improvements. Tomorrow you'll have a chance to make some final touches on your app before we submit them!

Assessment Opportunity

Create Performance Task Written Response Practice

In order to help prepare students for the new component of the AP Create Performance Task, there will be several opportunities throughout the curriculum to practice creating written responses aligned with the learning objectives that might be assessed on the exam.

Sample Prompt: *Describe the process you used to develop your program. Include any goals that you set, challenges or setbacks that you encountered, and solutions that worked for your team.*

Unit Journal Template

Day 5 (45 minutes)

Warm Up

Remarks

Your apps are almost done and soon will be ready to share with the world. Let's get back to work.

Teaching Tip

Get to the Project: Quickly move the class to the activity where the bulk of their time should be spent today.

Activity (38 minutes)

Group: Place students back in pairs with their project partner.

 **Distribute:** Make sure students have access to the [App Development Planning Guide](#)

Finish Building Apps

Remarks

For the next thirty minutes, you and your partner will work on finishing your app. Don't forget to consider the suggestions your classmates made. Check the rubric before submitting your app.

 **Code Studio:** Direct students to continue working on their app. When they are finished, they should hit "Submit" to turn in their app.

 3



Submit Your App

Remarks

Great work on your project. Before wrapping up, we will practice writing for the style of questions that are asked as part of the Create performance task on the AP exam. Respond to the question in Step 9 in your activity guide, and submit it along with your project.

 **Do This:** Complete the written response in Step 9 of the Project Guide.

Teaching Tip

Each student, in every classroom, will come to the written response with a different level of comfort, ability, and familiarity with writing for these style of questions. As such, consider offering one or more of the following supports:

Writing Support: Provide students with these sentence starters for guiding their response.

App Interface Support: For students who did not fully complete their app, direct them to the completed app in Level 1 in order to respond to the question

Wrap Up (7 minutes)

Remarks

Now that we are near the end of the unit, we are going to play CSP Vocabulary BINGO once again. You have been working hard on learning the definitions of all of the different vocabulary words throughout the unit, and now we will put your amazing memories to the test.

I will give each of you a BINGO card that has randomized vocabulary on it. When you hear the definition for a word that you have on your card, mark that square. If you have 4 in a row horizontally, vertically, or diagonally, yell, "BINGO!"

Are there any questions?

 **Do This:** Pass out a BINGO Card to all students. A printable PDF is located in the Resources section of this lesson plan.

 **Do This:** Use the [**CSP Bingo Generator**](#) for Unit Three in order to generate clues for the students.

 4

End of Unit Survey



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